Solved selected problems of General Relativity - Thomas A. Moore

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Chapter 21 - The Einstein Equation

Solution. **BOX 21.1** - Exercise 21.1.1.
Let us consider a LIF. We know that

$$\nabla_\delta R^{\gamma\sigma} = g^{\gamma\beta} g^{\sigma\nu} g^{\alpha\mu} \nabla_\delta R_{\alpha\beta\mu\nu}$$

If we set $\delta = \sigma$ then

$$\begin{aligned}\nabla_\sigma R^{\gamma\sigma} &= g^{\gamma\beta} g^{\sigma\nu} g^{\alpha\mu} \nabla_\sigma R_{\alpha\beta\mu\nu} \\ &= \frac{1}{2} g^{\gamma\beta} g^{\sigma\nu} g^{\alpha\mu} (\partial_\sigma \partial_\beta \partial_\mu g_{\alpha\nu} + \partial_\sigma \partial_\alpha \partial_\nu g_{\beta\mu} - \partial_\sigma \partial_\beta \partial_\nu g_{\alpha\mu} - \partial_\sigma \partial_\alpha \partial_\mu g_{\beta\nu})\end{aligned}$$

We can rename $\alpha \leftrightarrow \sigma$ and $\mu \leftrightarrow \nu$ since they are dummy indices so for the second term we see that

$$T_2 = g^{\alpha\mu} g^{\sigma\nu} \partial_\alpha \partial_\sigma \partial_\mu g_{\beta\nu}$$

Which is identical to the fourth term so they cancel out. Doing the same thing to the first term we see that

$$T_1 = g^{\alpha\mu} g^{\sigma\nu} \partial_\alpha \partial_\beta \partial_\nu g_{\sigma\mu}$$

But this is not the same as the third term so we cannot cancel them out, hence

$$\nabla_\sigma R^{\gamma\sigma} = \frac{1}{2} g^{\gamma\beta} g^{\sigma\nu} g^{\alpha\mu} (\partial_\sigma \partial_\beta \partial_\mu g_{\alpha\nu} - \partial_\sigma \partial_\beta \partial_\nu g_{\alpha\mu})$$

□

Solution. **BOX 21.2** - Exercise 21.2.1.
The second term of equation (21.30) is

$$+\nabla_\nu g^{\gamma\sigma} g^{\alpha\mu} g^{\beta\nu} R_{\alpha\beta\sigma\mu}$$

We know that $R_{\alpha\beta\sigma\mu} = +R_{\sigma\mu\alpha\beta}$ and that $R_{\sigma\mu\alpha\beta} = -R_{\mu\sigma\alpha\beta}$ because of the symmetries of the Riemann tensor, hence the second term of equation (21.30) can be written as

$$-\nabla_\nu g^{\gamma\sigma} g^{\alpha\mu} g^{\beta\nu} R_{\mu\sigma\alpha\beta}$$

In the same way for the third term, we have that

$$R_{\alpha\beta\nu\sigma} = R_{\nu\sigma\alpha\beta} = -R_{\nu\sigma\beta\alpha}$$

□

Solution. **BOX 21.2** - Exercise 21.2.2.
From equation (21.30) and applying the definition of Ricci tensor, we have that

$$\begin{aligned}
\nabla_\sigma g^{\gamma\sigma} R - \nabla_\nu g^{\gamma\sigma} g^{\alpha\mu} g^{\beta\nu} R_{\mu\sigma\alpha\beta} - \nabla_\mu g^{\gamma\sigma} g^{\alpha\mu} g^{\beta\nu} R_{\nu\sigma\beta\alpha} &= 0 \\
\nabla_\sigma g^{\gamma\sigma} R - \nabla_\nu g^{\gamma\sigma} g^{\beta\nu} R_{\sigma\beta} - \nabla_\mu g^{\gamma\sigma} g^{\alpha\mu} R_{\sigma\alpha} &= 0 \\
\nabla_\sigma g^{\gamma\sigma} R - \nabla_\nu R^{\gamma\nu} - \nabla_\mu R^{\gamma\mu} &= 0 \\
\nabla_\sigma g^{\gamma\sigma} R - 2\nabla_\sigma R^{\gamma\sigma} &= 0 \\
\frac{1}{2}\nabla_\sigma g^{\gamma\sigma} R - \nabla_\sigma R^{\gamma\sigma} &= 0
\end{aligned}$$

Where we contracted $R_{\sigma\alpha}$ and $R_{\sigma\beta}$ and we renamed the dummy indices $\nu \leftrightarrow \sigma$ and $\mu \leftrightarrow \sigma$. □

Solution. **BOX 21.3** - Exercise 21.3.1.
We know that $g_{\nu\sigma}g^{\mu\sigma} = \delta^{\mu}{}_{\nu}$ so if $\nu = \mu$ then we are summing the diagonal elements of $\delta^{\mu}{}_{\nu}$ (the trace), hence

$$\delta^{\mu}{}_{\mu} = 1 + 1 + 1 + 1 = 4$$

□

Solution. **BOX 21.3** - Exercise 21.3.2.
We know that $R = g_{\mu\nu}R^{\mu\nu}$ and that $T = g_{\mu\nu}T^{\mu\nu}$ then starting from equation (21.33) we see that

$$\begin{aligned} g_{\mu\nu}R^{\mu\nu} - \frac{1}{2}g_{\mu\nu}g^{\mu\nu}R + \Lambda g_{\mu\nu}g^{\mu\nu} &= \kappa g_{\mu\nu}T^{\mu\nu} \\ R - 2R + 4\Lambda &= \kappa T \\ -R + 4\Lambda &= \kappa T \end{aligned}$$

□

Solution. **P21.1**

(a) The Einstein equation in the Newtonian limit states that

$$\nabla^2\Phi = \frac{1}{2}\kappa\rho - \Lambda$$

If we consider the empty-space, then $\rho = 0$ and hence

$$\nabla^2\Phi = -\Lambda$$

Also, for this problem we know that

$$\nabla^2\Phi = \frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}\Phi}{\mathrm{d}r}\right)$$

Hence

$$\begin{aligned}\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}\Phi}{\mathrm{d}r}\right) &= -\Lambda\\ \frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}\Phi}{\mathrm{d}r}\right) &= -\Lambda r^2\end{aligned}$$

Therefore, by integration we see that

$$\begin{aligned}\int_0^r \frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}\Phi}{\mathrm{d}r}\right)\, dr &= -\Lambda\int_0^r r^2\, dr\\ r^2\frac{\mathrm{d}\Phi}{\mathrm{d}r} &= -\Lambda\frac{r^3}{3}\\ -\frac{\mathrm{d}\Phi}{\mathrm{d}r} &= \frac{\Lambda}{3}r\end{aligned}$$

(b) Let us require that

$$\frac{1}{3}\Lambda r \ll \frac{GM_\odot}{r^2}$$

If we consider a scale where $r \sim 10^{12}\ m$ and $GM_\odot \sim 1500\ m$ then

$$\frac{\Lambda}{8\pi G} \ll \frac{3GM_\odot}{8\pi G r^3} = \frac{3\cdot 1500}{8\pi\cdot 1477\cdot(10^{12})^3} = 1.21\times 10^{-37}\ \frac{\text{solar mass}}{m^3}$$

Hence

$$\frac{\Lambda}{8\pi G} \ll 2.41\times 10^{-7}\ \frac{kg}{m^3}$$

Where we used that $M_\odot = 2\times 10^{30}\ kg$.

□

Solution. **P21.2**
The Einstein tensor is defined as

$$G^{\mu\nu} = R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R = R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}g_{\alpha\beta}R^{\alpha\beta}$$

So if $R^{\mu\nu} = 0$ both terms become 0 and hence $G^{\mu\nu} = 0$.
On the other hand, if $G^{\mu\nu} = 0$ then

$$R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R = 0$$

Multiplying both sides of the equation by $g_{\mu\nu}$ we get that

$$\begin{aligned} g_{\mu\nu}R^{\mu\nu} - \frac{1}{2}g_{\mu\nu}g^{\mu\nu}R &= 0 \\ R - \frac{1}{2}4R &= 0 \\ R &= 0 \end{aligned}$$

Where we used that $g_{\mu\nu}g^{\mu\nu} = \delta^{\mu}_{\mu} = 4$ and that $R = g_{\mu\nu}R^{\mu\nu}$.
Therefore, since $R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R = 0$ we see that

$$R^{\mu\nu} = \frac{1}{2}g^{\mu\nu}R = 0$$

□

Solution. **P21.3**

a. Let us consider a perfect fluid in an arbitrary coordinate system, then

$$T^{\mu\nu} = (\rho_0 + p_0)u^\mu u^\nu + p_0 g^{\mu\nu}$$

And hence

$$\begin{aligned} T &= g_{\mu\nu}T^{\mu\nu} \\ &= g_{\mu\nu}(\rho_0 + p_0)u^\mu u^\nu + p_0 g_{\mu\nu}g^{\mu\nu} \\ &= -(\rho_0 + p_0) + 4p_0 \\ &= 3p_0 - \rho_0 \end{aligned}$$

Where we used that $g_{\mu\nu}u^\mu u^\nu = \boldsymbol{u} \cdot \boldsymbol{u} = -1$. Therefore

$$\begin{aligned} T^{\mu\nu} - \frac{1}{2}g^{\mu\nu}T &= (\rho_0 + p_0)u^\mu u^\nu + p_0 g^{\mu\nu} - \frac{1}{2}g^{\mu\nu}(3p_0 + \rho_0) \\ &= (\rho_0 + p_0)u^\mu u^\nu + \frac{1}{2}g^{\mu\nu}(\rho_0 - p_0) \end{aligned}$$

b. On a LOF where the fluid is at rest we have that

$$T^{\mu\nu} = \begin{bmatrix} \rho_0 & 0 & 0 & 0 \\ 0 & p_0 & 0 & 0 \\ 0 & 0 & p_0 & 0 \\ 0 & 0 & 0 & p_0 \end{bmatrix}$$

Also, $u^t = 1$, $u^i = 0$ for $i \in \{x, y, z\}$ and $g^{\mu\nu} = \eta^{\mu\nu}$. Then

$$\begin{aligned} T^{tt} - \frac{1}{2}\eta^{tt}T &= \rho_0 + p_0 - \frac{1}{2}(\rho_0 - p_0) = \frac{1}{2}(\rho_0 + 3p_0) \\ T^{xx} - \frac{1}{2}\eta^{xx}T &= \frac{1}{2}(\rho_0 - p_0) \\ T^{yy} - \frac{1}{2}\eta^{yy}T &= \frac{1}{2}(\rho_0 - p_0) \\ T^{zz} - \frac{1}{2}\eta^{zz}T &= \frac{1}{2}(\rho_0 - p_0) \end{aligned}$$

□

Solution. **P21.4**

The vacuum stress-energy tensor $T^{\mu\nu}_{vac}$ depends only on $g^{\mu\nu}$, in any LIF $g^{\mu\nu} = \eta^{\mu\nu}$ which components are constant, so $T^{\mu\nu}_{vac}$ is the same for every observer in a LIF. $\square$

Solution. **P21.5**
Let us consider a universe of two dimensions, where $R_{\mu\nu} = 0$.
In two dimensions, we know that there is only one independent component of the Riemann tensor and, it is related to the non-zero components as

$$R_{0101} = -R_{0110} = R_{1010} = -R_{1001}$$

The rest of the Riemann tensor components are 0.
We see that

$$\begin{aligned}
R_{00} &= g^{00}R_{0000} + g^{01}R_{0010} + g^{10}R_{1000} + g^{11}R_{1010} = g^{11}R_{1010} = g^{11}R_{0101} \\
R_{01} &= g^{00}R_{0001} + g^{01}R_{0011} + g^{10}R_{1001} + g^{11}R_{1011} = g^{10}R_{1001} = -g^{10}R_{0101} \\
R_{10} &= g^{00}R_{0100} + g^{01}R_{0110} + g^{10}R_{1100} + g^{11}R_{1110} = g^{01}R_{0110} = -g^{01}R_{0101} \\
R_{11} &= g^{00}R_{0101} + g^{01}R_{0111} + g^{10}R_{1101} + g^{11}R_{1111} = g^{00}R_{0101}
\end{aligned}$$

Given that $R_{\mu\nu} = 0$, if we assume that not all the metric components are 0 then must be that

$$R_{0101} = 0$$

Therefore, the only independent Riemann tensor component is 0, and hence the spacetime must be flat. □

Solution. **P21.6**
Let us consider a universe of three dimensions, where $R_{\mu\nu} = 0$.
In three dimensions, we know because of problem P19.2 that there are 6 independent components of the Riemann tensor and they are

$$R_{0101} \quad R_{0202} \quad R_{1212} \quad R_{0102} \quad R_{0212} \quad R_{0112}$$

If we take a LIF we see that

$$\begin{aligned} R_{00} &= g^{11}R_{1010} + g^{12}R_{1020} + g^{21}R_{2010} + g^{22}R_{2020} \\ &= g^{11}R_{0101} + (g^{12} + g^{21})R_{0102} + g^{22}R_{0202} \\ &= R_{0101} + R_{0202} \end{aligned}$$

Where in the last step we used that $g^{\mu\nu} = \eta^{\mu\nu}$ in a LIF. In the same way

$$\begin{aligned} R_{01} &= g^{22}R_{2021} = R_{2021} = R_{0212} \\ R_{02} &= g^{11}R_{1012} = R_{1012} = -R_{0112} \\ R_{10} &= R_{01} = R_{0212} \\ R_{11} &= g^{00}R_{0101} + g^{22}R_{2121} = R_{0101} + R_{1212} \\ R_{12} &= g^{00}R_{0102} = R_{0102} \\ R_{20} &= R_{02} = -R_{0112} \\ R_{21} &= R_{12} = R_{0102} \\ R_{22} &= g^{00}R_{0202} + g^{11}R_{1212} = R_{0202} + R_{1212} \end{aligned}$$

Since $R_{\mu\nu} = 0$ then we see that

$$R_{0212} = R_{0112} = R_{0102} = 0$$

But also we get that

$$R_{0101} + R_{0202} = 0 \quad R_{0101} + R_{1212} = 0 \quad R_{0202} + R_{1212} = 0$$

Then $R_{0101} = -R_{0202}$, $R_{0101} = -R_{1212}$ which implies that $R_{0202} = R_{1212}$ but also $R_{0202} + R_{1212} = 0$ then

$$R_{0202} = R_{1212} = 0$$

Replacing this into the first equation we see that $R_{0101} = 0$.
Therefore the Riemann tensor is 0 in the LIF, so the Riemann tensor is 0 in any coordinate system and hence the spacetime must be flat in vacuum. □

Solution. **P21.7**
The metric of a two dimensional space on a spherical surface is

$$g_{\mu\nu} = r^2 \begin{bmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{bmatrix} \qquad g^{\mu\nu} = \frac{1}{r^2} \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\sin^2\theta} \end{bmatrix}$$

and because of BOX 19.6 we know that

$$R_{\theta\theta} = 1 \qquad R_{\theta\phi} = R_{\phi\theta} = 0 \qquad R_{\phi\phi} = \sin^2\theta$$

and that $R = 2/r^2$. So

$$\begin{aligned} R^{\theta\theta} &= g^{\theta\alpha} g^{\theta\beta} R_{\alpha\beta} = \frac{1}{r^4} \\ R^{\theta\phi} &= R^{\phi\theta} = g^{\theta\alpha} g^{\phi\beta} R_{\alpha\beta} = 0 \\ R^{\phi\phi} &= g^{\phi\alpha} g^{\phi\beta} R_{\alpha\beta} = \frac{1}{r^4 \sin^2\theta} \end{aligned}$$

Let us compute the Einstein tensor as follows

$$\begin{aligned} G^{\theta\theta} &= R^{\theta\theta} - \frac{1}{2} g^{\theta\theta} R = \frac{1}{r^4} - \frac{1}{2}\frac{1}{r^2}\frac{2}{r^2} = 0 \\ G^{\theta\phi} &= R^{\theta\phi} - \frac{1}{2} g^{\theta\phi} R = 0 - 0 = 0 \\ G^{\phi\theta} &= R^{\phi\theta} - \frac{1}{2} g^{\phi\theta} R = 0 - 0 = 0 \\ G^{\phi\phi} &= R^{\phi\phi} - \frac{1}{2} g^{\phi\phi} R = \frac{1}{r^4 \sin^2\theta} - \frac{1}{2}\frac{1}{r^2 \sin^2\theta}\frac{2}{r^2} = 0 \end{aligned}$$

so we see that $G^{\mu\nu} = 0$. Since the Einstein equation is given by

$$G^{\mu\nu} = 8\pi G T^{\mu\nu}_{all}$$

Then we see that $T^{\mu\nu}_{all} = 0$ i.e. the stress-energy tensor is zero. □

Solution. **P21.8**
Let us consider the metric

$$ds^2 = -e^{2gx}dt^2 + dx^2 + dy^2 + dz^2$$

a. We know that always must happen that

$$\begin{aligned} \boldsymbol{u} \cdot \boldsymbol{u} &= -1 \\ g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} &= -1 \end{aligned}$$

Then, since the particle is at rest, we get that

$$\begin{aligned} -e^{2gx}\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)^2 &= -1 \\ \left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)^2 &= e^{-2gx} \\ \frac{\mathrm{d}t}{\mathrm{d}\tau} &= e^{-gx} \end{aligned}$$

On the other hand, the geodesic equation states that

$$0 = \frac{\mathrm{d}}{\mathrm{d}\tau}\left(g_{\mu\nu}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}\right) - \frac{1}{2}\partial_\mu g_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}x^\beta}{\mathrm{d}\tau}$$

Then the geodesic equation for $\mu = x$ gives us

$$\begin{aligned} 0 &= \frac{\mathrm{d}^2x}{\mathrm{d}\tau^2} + \frac{1}{2}\partial_x e^{2gx}\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)^2 \\ 0 &= \frac{\mathrm{d}^2x}{\mathrm{d}\tau^2} + ge^{2gx}\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)^2 \end{aligned}$$

Therefore, replacing $dt/d\tau$ we see that

$$\frac{\mathrm{d}^2x}{\mathrm{d}\tau^2} = -ge^{2gx}\left(e^{-gx}\right)^2 = -g$$

b. The geodesic equation for $\mu = t$ gives us

$$\begin{aligned} 0 &= \frac{\mathrm{d}}{\mathrm{d}\tau}\left(-e^{2gx}\frac{\mathrm{d}t}{\mathrm{d}\tau}\right) - 0 \\ 0 &= -2ge^{2gx}\frac{\mathrm{d}x}{\mathrm{d}\tau}\frac{\mathrm{d}t}{\mathrm{d}\tau} - e^{2gx}\frac{\mathrm{d}^2t}{\mathrm{d}\tau^2} \\ 0 &= \frac{\mathrm{d}^2t}{\mathrm{d}\tau^2} + 2g\frac{\mathrm{d}x}{\mathrm{d}\tau}\frac{\mathrm{d}t}{\mathrm{d}\tau} \end{aligned}$$

So we see that $\Gamma^t_{xt} = \Gamma^t_{tx} = g$.
From the geodesic equation for $\mu = x$ that we solved on part (a) we

also have that $\Gamma^x_{tt} = ge^{2gx}$.
For $\mu = y$ or $\mu = z$ the geodesic euqation gives us

$$\frac{\mathrm{d}^2 y}{\mathrm{d}\tau^2} = \frac{\mathrm{d}^2 z}{\mathrm{d}\tau^2} = 0$$

So all Christoffel symbols with y and z as superscript are 0.

c. We know that the Riemann tensor components are given by

$$\begin{aligned} R_{\alpha\beta\mu\nu} &= g_{\alpha\gamma} R^{\gamma}_{\beta\mu\nu} \\ &= g_{\alpha\gamma}(\partial_\mu \Gamma^\gamma_{\beta\nu} - \partial_\nu \Gamma^\gamma_{\beta\mu} + \Gamma^\gamma_{\mu\delta}\Gamma^\delta_{\beta\nu} - \Gamma^\gamma_{\nu\sigma}\Gamma^\sigma_{\beta\mu}) \end{aligned}$$

Given that there are only three non-zero Christoffel symbols then γ can only be x or t.
If $\alpha = t$ then γ needs to be t as well. Both Christoffel symbols Γ^t_{xt} and Γ^t_{tx} are constant so the first two terms in the Riemann tensor equation are zero, and must be that $\beta = x$ since α must be different from β. Finally, if $\mu = t$ and $\nu = x$ we see that

$$\begin{aligned} R_{txtx} &= g_{tt}(\Gamma^t_{t\delta}\Gamma^\delta_{xx} - \Gamma^t_{x\sigma}\Gamma^\sigma_{xt}) \\ &= -g_{tt}\Gamma^t_{xt}\Gamma^t_{xt} \\ &= e^{2gx} g^2 \end{aligned}$$

In the case $\mu = x$ and $\nu = t$ we know that $R_{txxt} = -R_{txtx}$.
If now $\alpha = x$ then $\beta = t$, then the only possible values for μ and ν are x and t. Suppose $\mu = x$ and $\nu = t$ then we get that

$$\begin{aligned} R_{xtxt} &= g_{xx}(\partial_x \Gamma^x_{tt} - \partial_t \Gamma^x_{tx} + \Gamma^x_{x\delta}\Gamma^\delta_{tt} - \Gamma^x_{t\sigma}\Gamma^\sigma_{tx}) \\ &= \partial_x \Gamma^x_{tt} - \Gamma^x_{tt}\Gamma^t_{tx} \\ &= 2g^2 e^{2gx} - g^2 e^{2gx} \\ &= g^2 e^{2gx} \end{aligned}$$

This is correct since we know that

$$R_{txtx} = -R_{xttx} = R_{xtxt}$$

So essentially there is only one nonzero independent component of the Riemann tensor.

d. We can compute the Ricci tensor by contracting the Riemann tensor over its first and third indices

$$R_{\beta\nu} = g^{\alpha\mu} R_{\alpha\beta\mu\nu}$$

We note that β and ν can be x or t. Suppose $\beta = t$ and $\nu = t$ then

$$R_{tt} = g^{\alpha\mu} R_{\alpha t\mu t} = g^{xx} R_{xtxt} = g^2 e^{2gx}$$

If $\beta = x$ and $\nu = t$ then

$$R_{xt} = g^{\alpha\mu} R_{\alpha x \mu t} = g^{tx} R_{txxt} = 0$$

Where we used that necessarily α needs to be equal to t and μ needs to be equal to x so $\alpha \neq \beta$ and $\mu \neq \nu$. The same happens when $\beta = t$ and $\nu = x$. Finally, if $\beta = \nu = x$ we get that

$$R_{xx} = g^{\alpha\mu} R_{\alpha x \mu x} = g^{tt} R_{txtx} = -e^{-2gx} g^2 e^{2gx} = -g^2$$

e. The vacuum Einstein equation states that $R_{\mu\nu} = 0$ then, must be that

$$R_{tt} = g^2 e^{2gx} = 0 \qquad\qquad R_{xx} = -g^2 = 0$$

Since e^{2gx} is never 0 no matter the value of x then must be that $g = 0$ so both equations hold.

f. Suppose g is nonzero inside a fluid.
Let us compute the components of the Ricci tensor with upper indices when we are in vacuum, then

$$\begin{aligned} R^{tt} &= g^{t\alpha} g^{t\beta} R_{\alpha\beta} = e^{-2gx} e^{-2gx} g^2 e^{2gx} = g^2 e^{-2gx} \\ R^{xx} &= g^{x\alpha} g^{x\beta} R_{\alpha\beta} = -g^2 \end{aligned}$$

We know from equation (21.14) that

$$R^{\mu\nu} = 8\pi G \left(T^{\mu\nu} - \frac{1}{2} g^{\mu\nu} T \right) + \Lambda g^{\mu\nu}$$

if we assume that $\Lambda = 0$ then we have that

$$\begin{aligned} R^{tt} &= g^2 e^{-2gx} = 8\pi G \left(T^{tt} + \frac{1}{2} e^{-2gx} T \right) \\ R^{xx} &= -g^2 = 8\pi G \left(T^{xx} - \frac{1}{2} T \right) \\ R^{yy} &= 0 = 8\pi G \left(T^{yy} - \frac{1}{2} T \right) \\ R^{zz} &= 0 = 8\pi G \left(T^{zz} - \frac{1}{2} T \right) \end{aligned}$$

On the other hand, we know that $T = g_{\mu\nu} T^{\mu\nu}$ so

$$T = -e^{2gx} T^{tt} + T^{xx} + T^{yy} + T^{zz}$$

from the equations of R^{yy} and R^{zz} we see that

$$T^{yy} = \frac{T}{2} \quad \text{and} \quad T^{zz} = \frac{T}{2}$$

Replacing this on the equation for T we get that

$$\begin{aligned} T &= -e^{2gx}T^{tt} + T^{xx} + T \\ 0 &= -e^{2gx}T^{tt} + T^{xx} \\ T^{xx} &= e^{2gx}T^{tt} \end{aligned}$$

If we add the equations for R^{tt} and R^{xx} we get that

$$\begin{aligned} g^2 - g^2 &= 8\pi G e^{2gx}\left(T^{tt} + \frac{1}{2}e^{-2gx}T\right) + 8\pi G\left(T^{xx} - \frac{1}{2}T\right) \\ 0 &= 8\pi G\left(e^{2gx}T^{tt} + \frac{1}{2}T + T^{xx} - \frac{1}{2}T\right) \\ 0 &= 8\pi G\left(e^{2gx}T^{tt} + T^{xx}\right) \\ T^{xx} &= -e^{2gx}T^{tt} \end{aligned}$$

Joining this result with the previous one we see that $T^{xx} = T^{tt} = 0$. So from the equation for R^{xx} we have that

$$\begin{aligned} -g^2 &= 8\pi G\left(0 - \frac{T}{2}\right) \\ \frac{T}{2} &= \frac{8\pi G}{g^2} \end{aligned}$$

Therefore

$$T^{zz} = T^{yy} = \frac{8\pi G}{g^2}$$

If we considered a LIF where our perfect fluid is at rest with these stress-energy components we would see that the fluid has zero energy density ρ_0, is making zero pressure in the x direction but it's making a non-zero pressure in the y and z direction, this does not make sense since the pressure must be the same in any direction, we cannot have a prefered direction.

□